



Geopolitical Risk in a Concentrated Supply Chain:

NVIDIA's Dependence on Taiwan

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Executive Summary

NVIDIA's rapid growth in AI and data center computing has increased its reliance on advanced semiconductor manufacturing, particularly in Taiwan. This report evaluates whether that dependence exposes the company to material geopolitical risk. By combining global supply chain data, industry trends, and firm-level disclosures, the analysis finds that semiconductor production is increasingly concentrated in Taiwan and the broader Asia-Pacific region.

While NVIDIA acknowledges these risks in formal filings, they are less emphasized in investor-facing communication. Overall, the company's reliance on Taiwan translates into structural exposure to potential supply disruptions, regulatory constraints, and long-term industry fragmentation.

Background

The global semiconductor supply chain is highly concentrated, with Taiwan accounting for a dominant share of advanced chip manufacturing and a disproportionate share of high-value semiconductor production¹. Taiwan produces over 60% of global semiconductors and more than 90% of advanced chips, making it a critical node in global technology production. This concentration creates a structural vulnerability, as disruptions to Taiwan's semiconductor industry would have widespread impacts across industries reliant on advanced computing and AI infrastructure.

At the same time, rising geopolitical tensions between China and Taiwan, characterized by increasing military activity, economic pressure, and the potential for blockade or conflict, have heightened concerns over supply chain stability and the risk of disruption to production and global shipping². A disruption scenario could affect a large share of global semiconductor output and severely constrain access to critical technologies. Together, these factors position the semiconductor supply chain as a key point of global economic and strategic vulnerability.

As a fabless semiconductor company reliant on Taiwan-based manufacturers such as TSMC, NVIDIA operates within and is directly exposed to these structural vulnerabilities in the global supply chain.

Company Summary

NVIDIA is a leading accelerated computing company specializing in Graphics Processing Unit (GPU)s, AI Infrastructure, and high-performance computing platforms. Originally focused on gaming graphics, it has evolved into a dominant provider of data center and AI solutions, with strong demand from cloud providers and enterprises. Its integrated hardware software ecosystem, centered on Compute Unified Device Architecture (CUDA), creates high switching costs and supports a broad range of applications.

Operating a fabless model, NVIDIA outsources manufacturing to partners like Taiwan Semiconductor Manufacturing Company (TSMC) while focusing on design and software. This strategy enables scalability but exposes the company to global supply chain risks in a competitive, rapidly evolving semiconductor industry.

1. Organisation for Economic Co-operation and Development (OECD). (2023). *Global semiconductor value chains: A strategic perspective*. OECD Publishing. <https://www.oecd.org>
2. Center for Strategic and International Studies (CSIS). (2023). *Semiconductor supply chains and the Taiwan risk*. <https://www.csis.org>

Motivation

NVIDIA faces significant geopolitical risk due to its reliance on a globally distributed highly concentrated semiconductor supply chain and increasingly regulatory intervention in AI technologies. Its fabless model depends heavily on third-party manufacturers in Asia, particularly Taiwan, making it structurally exposed to disruptions stemming from China-Taiwan tensions. At the same time, expanding U.S. export controls on advanced AI chips are constraining market access, especially in China, while accelerating global semi-conductor fragmentation. These risks are not cyclical but structural, with potential long-term implications for NVIDIA's supply continuity, revenue growth, and competitive position include:

- **Taiwan Supply Chain Disruption**—Heavy reliance on TSMC and Asia-Pacific supply chains for advanced chip production. A Taiwan conflict could halt production of advanced GPUs and severely disrupt supply.
- **U.S. Export Controls & Market Access Restrictions**—Expanding U.S. restrictions on AI chips limit NVIDIA's ability to sell into China and other regions. This reduces total addressable market, introduces regulatory uncertainty, and risks permanent loss of key customers.
- **Global Semiconductor Fragmentation & Competitive Substitution**—Geopolitical Tensions are accelerating regional self-sufficiency efforts (China, U.S.), incentivizing customers to adopt non-U.S. alternatives. Over time, this creates erosion of NVIDIA's market share and ecosystem dominance.

These three risks together capture the core exposure: supply shock, demand restriction, and long-term competitive displacement.

Research Question

This project evaluates whether NVIDIA's reliance on Taiwan-based semiconductor manufacturing exposes the company to material geopolitical risk. Specifically, the analysis focuses on identifying the extent of NVIDIA's dependence on third-party operational vulnerability to answer the question:

How does NVIDIA's reliance on Taiwan translate into financial and operational risk under potential geopolitical disruption scenarios?

Data Methods

The project employs a structured text-processing framework designed to convert narrative corporate disclosures into standardized representations of firm-level risk communication. The approach emphasizes consistency, reproducibility, and the creation of analytical units that can support subsequent quantitative analysis.

- **Supply Chain Analysis (ICIO)**—Uses ICIO data and industry codes to calculate Taiwan's share of global semiconductor value-added.

- **Industry Trend Analysis**—Uses WSTS data to analyze regional semiconductor billings and identify concentration in Asia-Pacific.
- **Text Processing and Feature Construction**—Extracts and cleans NVIDIA’s disclosures, then applies keyword frequency and normalization to measure risk-related language.
- **Sentiment and Comparative Analysis:** Uses sentiment scores and keyword data to compare how risk is communicated across documents.

Sources

Annual Report (Form 10-K)

The Form 10-K is a comprehensive, legally mandated annual filing that contains detailed narrative disclosures on firm-level risks, operational dependencies, regulatory exposure, and strategic vulnerabilities. It provides long-form, standardized textual data that serves as the primary foundation for analyzing how firms formally articulate risk in a consistent, year-over-year format.

Earnings Call Transcript (Q3 2025)

The Q3 2025 earnings call transcript contains conversational, unscripted dialogue between executives and analysts. This source provides qualitative narrative data that reflects managerial emphasis, tone, and forward-looking concerns. It complements the formal language of filings by revealing how firms verbally frame risks in real time.

OECD Inter-Country Input-Output Tables (ICIO)

The ICIO dataset is a global economic input-output matrix that quantifies cross-border production linkages and value-added flows across industries and countries. It provides structured numerical data that contextualizes firm-level disclosures within broader global supply-chain dependencies, enabling analysis of structural exposure beyond the narrative text.

Taiwan Domestic Input-Output Table (2022)

The Taiwan domestic input-output table provides sector-level detail on Taiwan’s internal production structure, including the semiconductor sector. It offers granular numerical data on domestic value-added composition, upstream and downstream linkages, and internal supply-chain concentration. This dataset complements the ICIO tables by supplying country-specific structural information not available in global matrices.

WSTS Historical Semiconductor Billings

The WSTS dataset contains monthly and annual semiconductor billing data across major product categories. It provides industry-level time-series data that captures cyclical demand patterns and macro-industry dynamics. This information helps distinguish between structural geopolitical risks and cyclical fluctuations when interpreting firm-level disclosures.

Data Preparation

SQL Database Integration

All datasets, including OECD ICIO tables, WSTS semiconductor billing data, and the Taiwan domestic input-output table, are imported and stored within a relational SQL database. This

enables efficient querying, aggregation, and consistent data management across large, multi-country datasets.

Text Data Preparation

The text data preparation process converts unstructured corporate disclosures into a standardized format suitable for systematic analysis. Firm-level disclosures from NVIDIA's 10-K and earnings call transcript are first extracted from PDF format using "*pdfplumber*", converting each document into raw text. The text undergoes preprocessing and normalization, involving the removal of URLs, non-alphabetic characters, and excess whitespace to standardize the corpus for analysis.

These preprocessing steps ensure that the textual data is consistent, comparable, and structured for subsequent keyword and sentiment analysis.

Analysis

Economic Context: Taiwan's Role in Semiconductors

Identify Relevant Semiconductor Industries

Industry classification codes from the ICIO dataset are filtered and used to isolate sectors relevant to semiconductor production, particularly:

- C26: Computer, Electronic, and Optical Products
- C27: Electrical Equipment

These industry codes serve as proxies for semiconductor-related activity, allowing the analysis to systematically extract relevant portions of global value-added rather than relying on ambiguous or manual classification.

Measuring Taiwan's Share of Global Semiconductor Production

SQL queries are used to calculate total global semiconductor value-added and isolate Taiwan's contribution within the same set of semiconductor-related industries. Taiwan's share is then computed as a proportion of global value-added and tracked over time to identify trends in geographic concentration.

Semiconductor value-added is calculated by aggregating ICIO data across industry codes C26–C28, which serve as proxies for semiconductor-related production. Global and Taiwan-specific values are computed separately, allowing Taiwan's contribution to be expressed as a share of total global value-added.

The results show that Taiwan accounts for approximately 7.23% of global semiconductor value-added in 2022. This is economically significant because it indicates that a single economy is responsible for a substantial share of value creation in a strategically critical industry. Given Taiwan's relatively small size in the global economy, this level of concentration highlights its outsized importance in semiconductor production and suggests a high degree of dependency within the global supply chain.

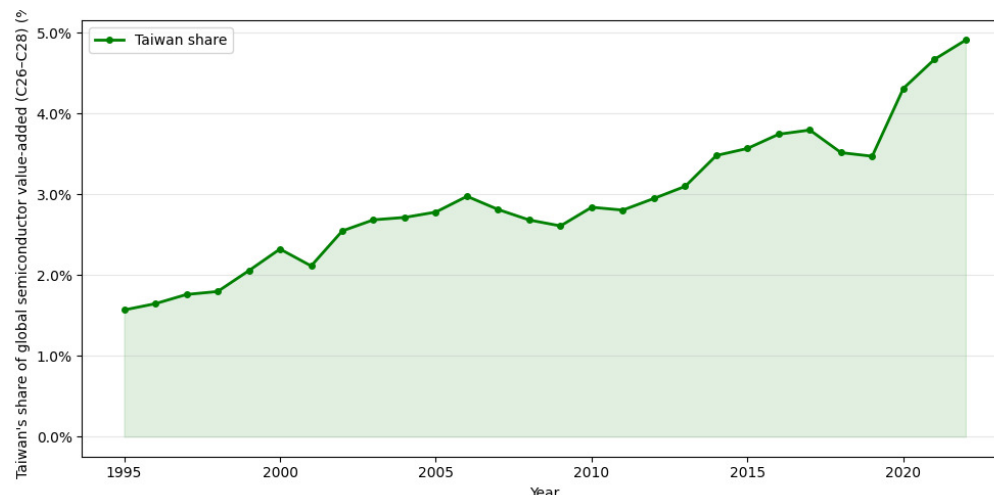
The analysis extends this calculation across multiple years, aligning global and Taiwan value-added by year to construct a time series of Taiwan's share. This enables evaluation of whether semiconductor production is becoming more geographically concentrated over time.

Table 1: Global and Taiwan Semiconductor Value-Added and Share (2018–2022)

	Year	global_va	tw_n_va	tw_n_share
0	2022	6718561.832	330023.484	0.049121
1	2021	6749301.960	315469.012	0.046741
2	2020	5826977.374	251344.055	0.043135
3	2019	5868608.109	203839.901	0.034734
4	2018	5996295.220	210956.987	0.035181

This step converts absolute production values into a relative measure of concentration, enabling direct comparison of Taiwan's importance within the global semiconductor supply chain over time. Tracking the ratio longitudinally is important because it shows whether Taiwan's role is stable or increasing. If Taiwan's share rises over time, it suggests that global semiconductor production is becoming more geographically concentrated rather than more diversified, which strengthens the argument that geopolitical disruption in Taiwan would create systemic rather than localized risk.

Figure 1: Taiwan Contribution to Global Semiconductor Production (1995–2022)



The table shows that Taiwan's share of global semiconductor value-added has increased steadily over time, rising from approximately 1.5% in 1995 to over 5% in 2022. This upward trend indicates growing geographic concentration in semiconductor production, particularly in high-value segments.

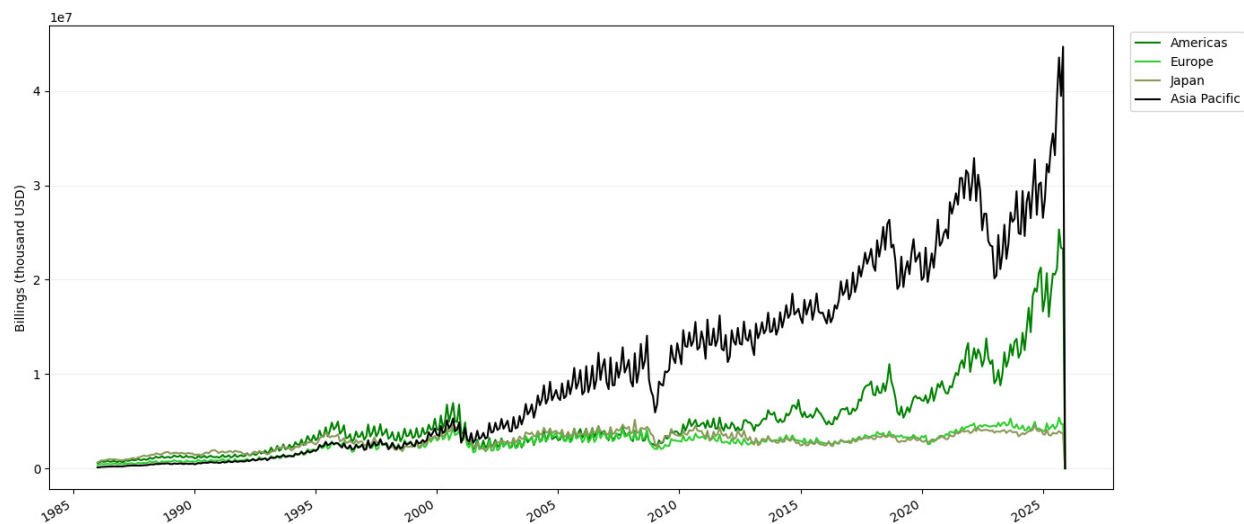
As a result, the global supply chain has become increasingly dependent on Taiwan, amplifying the potential impact of geopolitical disruption. This concentration reinforces the structural nature of supply risk and highlights the vulnerability of firms, such as NVIDIA, that rely on Taiwan-based manufacturing.

Industry Trend Analysis Using WSTS Data

To contextualize NVIDIA's geopolitical exposure within broader industry dynamics, WSTS semiconductor billing data is used to analyze long-run trends in global semiconductor activity and its geographic distribution. By examining regional billings over time, the analysis identifies shifts in where semiconductor production is concentrated and whether industry growth has become increasingly localized in specific regions.

WSTS semiconductor billing data is reshaped into a time-series format and aggregated by region to compare growth patterns across major markets. By examining differences in growth trajectories, the analysis identifies whether semiconductor activity is becoming increasingly concentrated in specific regions.

Figure 2: Regional Semiconductor Billings by Region (1986–2025)



The figure shows that Asia-Pacific has experienced significantly more rapid growth in semiconductor billings compared to other regions. While the Americas, Europe, and Japan display more gradual increases and relatively flatter trends, Asia-Pacific exhibits a much steeper upward trajectory, particularly in the post-2010 period. This indicates that not only is semiconductor activity concentrated in Asia, but it is also expanding at a faster rate than in other regions. Although all regions reflect cyclical fluctuations typical of the semiconductor industry, the magnitude and acceleration of growth in Asia-Pacific suggest a structural shift rather than temporary divergence. This widening gap implies that an increasing share of global semiconductor production and value is being captured by Asia over time.

From a risk perspective, the combination of higher concentration and faster growth amplifies exposure to regional disruptions. As semiconductor production becomes both more geographically concentrated and increasingly dominated by Asia-Pacific, firms such as NVIDIA that rely on this regional ecosystem face heightened vulnerability to geopolitical shocks.

Firm-Level Risk Disclosure Analysis

To complement the macro- and industry-level analysis, firm-level disclosures from NVIDIA's FY2026 10-K and Q3 earnings call transcript are analyzed to assess how the company communicates and frames geopolitical and supply chain risks. The analysis applies a structured text-processing approach, including keyword frequency analysis and sentiment analysis, to systematically evaluate differences in risk communication across documents.

Keyword Analysis

A set of predefined keywords related to geopolitical and supply chain risk, including “China,” “risk,” “geopolitical,” “supply,” and “manufacturing,” is used to quantify the frequency of risk-related language in each document.

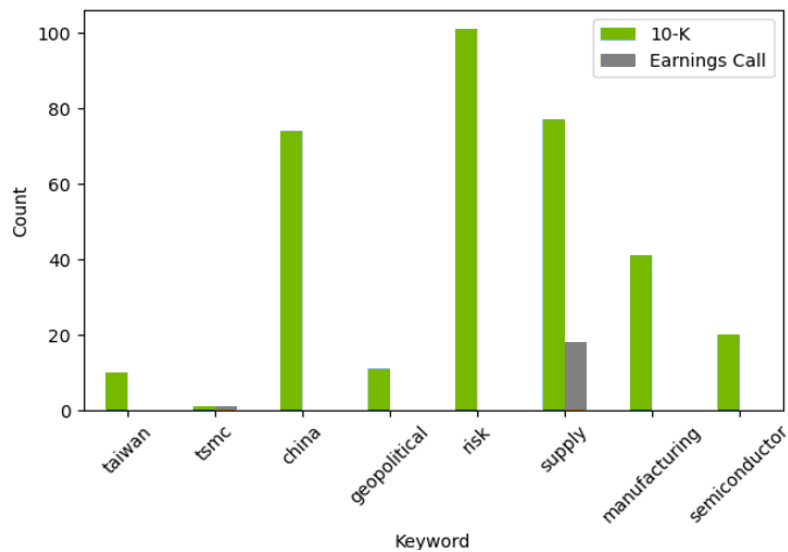
Firm-level disclosures are extracted from PDF format and converted into raw text, enabling systematic analysis of how NVIDIA communicates risk across different reporting contexts. The extracted text is normalized through preprocessing to remove noise and standardize formatting, ensuring that differences in keyword frequency reflect substantive content rather than inconsistencies in text structure. Keyword frequency analysis is applied to quantify the extent to which risk-related terms appear in each document, providing a measurable proxy for the emphasis placed on geopolitical and supply chain risks.

Table 2: Keyword Frequency of Risk-Related Terms in NVIDIA 10-K and Earnings Call

	Keyword	10-K	Earnings Call
0	taiwan	10	0
1	tsmc	1	1
2	china	74	0
3	geopolitical	11	0
4	risk	101	0
5	supply	77	18
6	manufacturing	41	0
7	semiconductor	20	0

The results show that the 10-K contains significantly higher mentions of these terms, with “risk” appearing 101 times and “China” 74 times, while these terms are largely absent in the earnings call.

Figure 3: Keyword Frequency Comparison: NVIDIA 10-K vs Earnings Call



This suggests that NVIDIA formally acknowledges geopolitical and supply chain risks in regulatory disclosures, but does not emphasize these risks in public-facing management discussions.

Additionally, while "supply" is mentioned in both sources, it appears more frequently in the 10-K and is discussed in the earnings call primarily in an operational context rather than as a geopolitical concern. References to Taiwan and TSMC are minimal, indicating that NVIDIA does not explicitly highlight its geographic manufacturing dependence in investor communications.

Normalized Keyword Analysis

To control for differences in document length, keyword counts are normalized by total word count. To ensure comparability across documents of different lengths, keyword frequencies are normalized by total word count, allowing for relative rather than absolute comparisons of emphasis.

Table 3: Raw and Normalized Keyword Frequencies in NVIDIA 10-K and Earnings Call

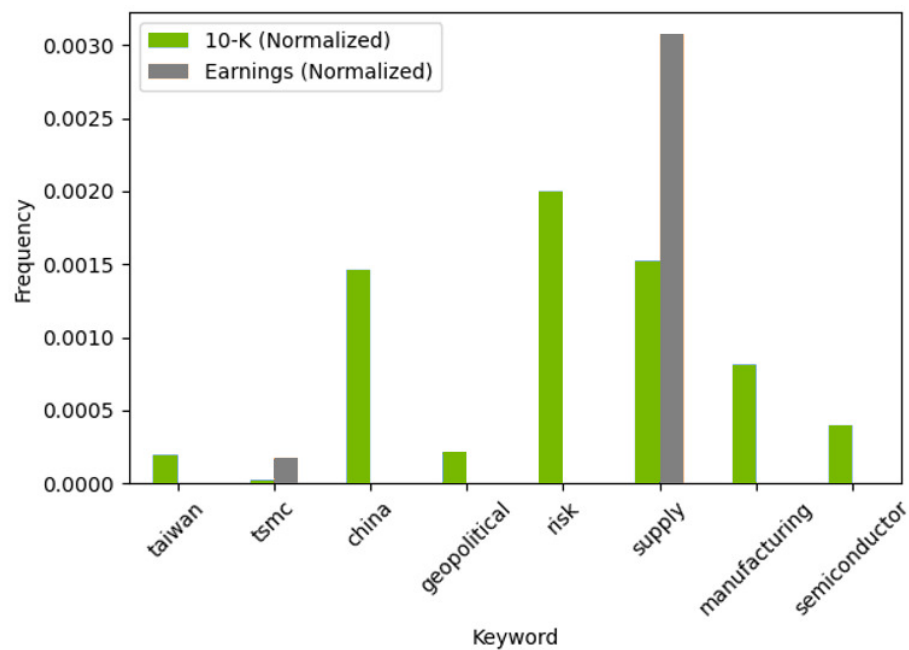
	Keyword	10-K (Raw)	Earnings (Raw)	10-K (Normalized)	Earnings (Normalized)
0	taiwan	10	0	0.000198	0.000000
1	tsmc	1	1	0.000020	0.000171
2	china	74	0	0.001463	0.000000
3	geopolitical	11	0	0.000218	0.000000
4	risk	101	0	0.001997	0.000000
5	supply	77	18	0.001523	0.003071
6	manufacturing	41	0	0.000811	0.000000
7	semiconductor	20	0	0.000396	0.000000

After adjusting for document length, the differences in keyword frequency remain significant. The 10-K continues to show a much higher relative frequency of terms such as "risk," "China," and "geopolitical," confirming that these themes are structurally emphasized in formal disclosures.

In contrast, the earnings call contains virtually no references to these terms, even after normalization. Instead, the term "supply" appears more frequently in the earnings call on a normalized basis, suggesting that NVIDIA discusses supply chain issues in an operational context rather than framing them as geopolitical risks.

These findings reinforce the conclusion that NVIDIA acknowledges geopolitical exposure in regulatory filings while avoiding explicit discussion of these risks in investor-facing communication.

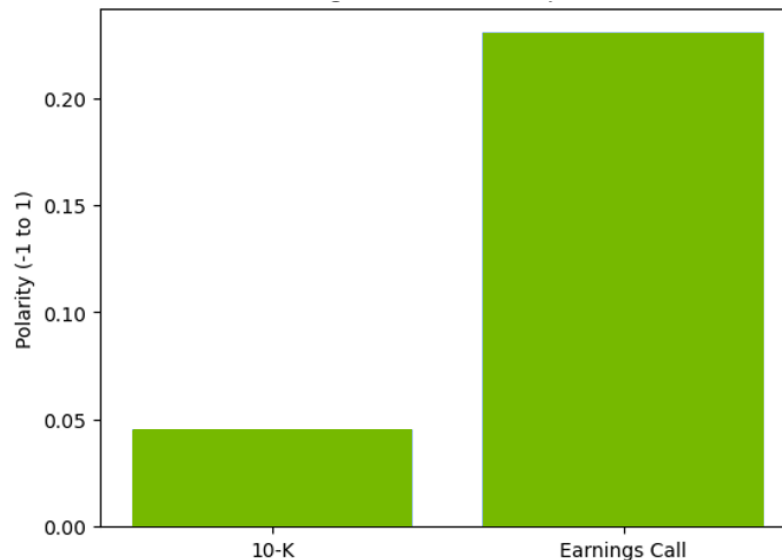
Figure 4: Normalized Keyword Frequency Comparison



Sentiment Analysis

Sentiment analysis is conducted to quantify the overall tone of each document, enabling comparison of whether risk is framed in a neutral, negative, or positive manner across different communication contexts.

Figure 5: Sentiment Comparison: NVIDIA 10-K vs Earnings Call



The earnings call exhibits a significantly more positive sentiment (0.23) compared to the 10-K filing (0.045), indicating that management adopts a more optimistic tone when communicating with investors.

In contrast, the 10-K maintains a more neutral tone, consistent with its role as a formal disclosure document outlining risks and uncertainties. This difference in sentiment reinforces earlier findings that NVIDIA emphasizes geopolitical and supply chain risks in regulatory filings while presenting a more positive and forward-looking narrative in earnings calls.

Together, the keyword and sentiment analyses suggest a divergence between how NVIDIA communicates risk in formal disclosures versus investor-facing discussions.

Results

The analysis combines global supply chain data, industry trends, and firm-level disclosures to evaluate NVIDIA's exposure to geopolitical risk.

At the global level, ICIO data shows that Taiwan accounts for approximately 7.23% of global semiconductor value-added in 2022, with its share increasing steadily over time. This upward trend indicates growing geographic concentration in semiconductor production, particularly in high-value segments, reinforcing Taiwan's central role in the global supply chain.

At the industry level, WSTS semiconductor billing data reveals that Asia-Pacific has experienced significantly faster growth than other regions, with a widening gap relative to the Americas, Europe, and Japan. While all regions exhibit cyclical fluctuations, the sustained divergence suggests a structural shift toward regional concentration in Asia, rather than temporary variation.

At the firm level, text analysis of NVIDIA's 10-K and earnings call highlights a divergence in risk communication. The 10-K contains significantly higher mentions of risk-related and geopolitical terms, including "risk" (101 mentions) and "China" (74 mentions), while these terms are largely

absent in the earnings call. Normalized results confirm this pattern, indicating that geopolitical risks are emphasized in formal disclosures but not in investor-facing communication.

Sentiment analysis further supports this distinction, showing that the earnings call exhibits a more positive tone (0.23) compared to the 10-K (0.045). This suggests that NVIDIA frames its messaging differently depending on the context, maintaining neutrality in regulatory filings while adopting a more optimistic tone in investor communications.

Taken together, these results show a consistent pattern across all levels of analysis:

Semiconductor production is increasingly concentrated in Asia and Taiwan, and NVIDIA is both structurally exposed to this concentration and aware of the associated risks.

Conclusion

NVIDIA's reliance on Taiwan-based semiconductor manufacturing exposes the company to material geopolitical risk, with direct implications for its financial and operational stability. Evidence from global supply chain data shows that semiconductor value-added is increasingly concentrated in Taiwan, while industry trends indicate that production growth is heavily centered in the Asia-Pacific region. These structural patterns imply that a disruption in Taiwan would have widespread effects on global semiconductor supply and directly constrain NVIDIA's production capacity.

Firm-level analysis confirms that NVIDIA acknowledges these risks in formal disclosures, particularly in its 10-K filings. However, the limited emphasis on geopolitical risk in earnings call communications suggests that these risks are not fully reflected in investor-facing narratives.

Overall, NVIDIA's dependence on Taiwan translates into tangible risk through potential supply disruptions, reduced production capacity, and exposure to regulatory constraints such as export controls. Although mitigation strategies exist, the concentration of advanced semiconductor manufacturing in Taiwan limits the company's ability to diversify away from this exposure, making the risk both structural and persistent.

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